

1. Administrative & Study Identification

Full Study Title: A Phase 1b/2 Open-label, Multicenter Study to Evaluate the Safety and Efficacy of Raludotatug Deruxtecan With or Without Other Anticancer Investigational Agents in Participants With High-grade Serous Epithelial Ovarian, Primary Peritoneal, or Fallopian Tube Cancer Who Have Relapsed After Prior Platinum-based Chemotherapy **Short Title/Acronym:** REJOICE-Ovarian02 / MK-5909-003 **Protocol Number:** MK-5909-003 **NCT Number:** NCT06843447 **Trial Status:** Recruiting **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Raludotatug Deruxtecan (R-DXd / MK-5909 / DS-6000) **Phase of Development:** Phase 1b/2 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, adverse events, and Dose-Limiting Toxicities (DLTs) of R-DXd when combined with other anticancer agents (carboplatin, paclitaxel, bevacizumab, or pembrolizumab) to determine the Recommended Phase 2 Dose (RP2D). **Secondary Objectives:** To evaluate the efficacy of the treatment regimens as measured by Objective Response Rate (ORR), Duration of Response (DOR), and Progression-Free Survival (PFS).

2.2 Background & Rationale To address the critical unmet medical need in treating recurrent high-grade serous epithelial ovarian cancer, primary peritoneal cancer, and fallopian tube cancer. Recurrent high-grade serous ovarian cancer is invariably fatal, and there are limited options once patients relapse after prior platinum-based chemotherapy. Raludotatug deruxtecan (R-DXd) is an antibody-drug conjugate (ADC) targeting CDH6, delivering a potent cytotoxic payload directly to tumor cells. This study aims to determine if R-DXd is safe and effective when administered alongside standard chemotherapy or targeted therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Part 1: Dose Escalation; Part 2: Efficacy Expansion) **Control Type:** Single-arm/Non-randomized (for each combination cohort) **Blinding:** Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Target \$N\$: 280 participants
Number of Sites: Multicenter **Estimated Study Start:** 15-Apr-2025
Estimated Primary Completion: 27-Mar-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Pathologically documented diagnosis of high-grade serous epithelial ovarian cancer, primary peritoneal cancer, or fallopian tube cancer. 2. Has measurable disease per Response Evaluation Criteria In Solid Tumors version 1.1 (RECIST 1.1). 3. Has provided tumor tissue from a core or excisional biopsy of a tumor lesion not previously irradiated. 4. Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 7 days before allocation. 5. **Cohort A-1 (Arm 2 and Arm 3):** Relapsed disease after 1 to 3 prior lines of therapy and radiographic evidence of disease progression ≥ 6 months (≥ 180 days) after the last dose of platinum-based therapy (platinum-sensitive disease). 6. **Cohort B-1 and B-2:** Relapsed disease after 1 to 3 prior lines of therapy and radiographic evidence of disease progression < 6 months (< 180 days) after the last dose of platinum-based therapy (platinum-resistant disease) and is a candidate for bevacizumab treatment. 7. **Cohort C-1 and Cohort D:** Relapsed disease after 1 prior line of therapy, radiographic evidence of disease progression ≥ 6 months (≥ 180 days) after the last dose of platinum-based therapy (platinum-sensitive disease) and progressed during prior treatment with PARPi in the first-line setting. 8. Human immunodeficiency virus (HIV)-infected participants must have well-controlled HIV on anti-retroviral therapy. 9. Participants who are hepatitis B surface antigen positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy for at least 4 weeks and have undetectable HBV viral load prior to allocation. 10. Participants with a history of hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable at screening.

4.2 Exclusion Criteria 1. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 2. Has a history of (noninfectious) pneumonitis or interstitial lung disease (ILD) that required steroids, or has current pneumonitis/ILD. 3. Has received prior radiotherapy within 2 weeks of the start of study intervention, or has radiation-related toxicities requiring corticosteroids. 4. Has an active infection requiring systemic therapy. 5. Has any of the following within 6 months before allocation: cerebrovascular accident, transient ischemic attack, or other arterial thromboembolic event. 6. Has uncontrolled or significant cardiovascular disease. 7. Has clinically severe pulmonary compromise resulting from intercurrent pulmonary illnesses including, but not limited to, any underlying pulmonary disorder, and any autoimmune, connective tissue, or inflammatory disorders with potential pulmonary involvement, or prior pneumonectomy. 8. Has $\geq$ Grade 2 peripheral neuropathy. 9. Has received prior treatment with cadherin-6-targeted agents. 10. Has received prior systemic anticancer therapy including investigational agents within 4 weeks or 5 half-lives (whichever is shorter) before allocation.

11. Receives chronic steroid treatment. 12. Has known additional malignancy that is progressing or has required active treatment within the past 3 years. 13. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Escalating doses of raludotatug deruxtecan (R-DXd) in combination with carboplatin, paclitaxel, bevacizumab, or pembrolizumab on Day 1 of every 3-week cycle. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Maximum of six 3-week cycles for carboplatin (approx. 4 months), while R-DXd is administered until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: 5-HT3 Serotonin Receptor Antagonists, NK-1 receptor antagonists, and short-term corticosteroid premedication to prevent hypersensitivity. **Prohibited:** Chronic systemic steroid treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Biopsy Collection, ECG, ECHO, Viral Labs
Baseline	Day 0	Cohort Assignment, First Dose Infusion
Treatment Cycles	Every 3 Weeks	IV Infusion of study drugs, Safety Labs, DLT Monitoring
End of Study	Variable	Final safety assessment, Exit Interview, Follow-up for PFS

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Part 1:** Number of Participants Who Experience a Dose-Limiting Toxicity (DLT) Per Common Terminology Criteria for Adverse Events, Version 5.0 (CTCAE v5.0). 2. **Part 1:** Number of Participants with One or More Adverse Events (AEs). 3. **Part 1:** Number of Participants who Discontinue Study Intervention Due to an AE.

7.2 Secondary Endpoints 1. **Part 1:** Objective Response Rate (ORR). 2. **Part 2:** Duration of Response (DOR). 3. **Part 2:** Progression-free Survival (PFS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical examinations, blood sampling, and continuous CTCAE v5.0 safety monitoring to identify AEs and track discontinuations due to AEs. **Serious Adverse Events (SAEs):** Specific monitoring for drug-induced ILD/pneumonitis, thrombocytopenia, severe pulmonary compromise, and neutropenia. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by the sponsor to determine dose-escalation progressions based on observed DLTs.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints; Safety Set for all AE/SAE and DLT evaluations. **Sample Size Justification:** N=280 structured across Phase 1 dose escalation and Phase 2 expansion cohorts. **Handling of Missing Data:** Standard survival analysis metrics and censoring for PFS and DOR endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous safety monitoring during dose-escalation phases and standardized central radiographic review. **Audit Trail:** eCRF locking, tracking of protocol deviations, and Blinded Independent Central Review (BICR) of imaging data.

11. Study Identifiers & Governance

Primary ID: MK-5909-003 **Posting ID:** 803265315070577003 **Registry Number:** NCT06843447 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of Raludotatug Deruxtecan in People With Ovarian Cancer (MK-5909-003) **Official Regulatory Title:** A Phase 1b/2 Open-label, Multicenter Study to Evaluate the Safety and Efficacy of Raludotatug Deruxtecan With or Without Other Anticancer Investigational Agents in Participants With High-grade Serous Epithelial Ovarian, Primary Peritoneal, or Fallopian Tube Cancer Who Have Relapsed After Prior Platinum-based Chemotherapy

12. Clinical Framework (Oncology Specific)

Primary Condition: Recurrent Ovarian Cancer (High-grade Serous Epithelial Ovarian, Primary Peritoneal, or Fallopian Tube Cancer) **Diagnosis Threshold:** Radiographic evidence of disease progression after 1 to 3 prior lines of therapy (including specific

requirements for platinum-sensitive and platinum-resistant disease) **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) Therapy **Optimization Target:** Recommended Phase 2 Dose (RP2D) determination via safety and DLT profiling

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for relapsed disease **Education Protocol (HCP):** Dose escalation monitoring, DLT tracking, AE evaluation, and CTCAE grading **Education Protocol (Patient):** IV Infusion schedule adherence and adverse event awareness **Quality Audit Mechanism:** Tumor imaging evaluations per RECIST 1.1 guidelines

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Day 1 of every 3-week cycle **Remote Monitoring Frequency:** Continuous safety follow-up **Checklist Review Interval:** Regular tumor imaging assessments per RECIST 1.1 evaluation schedule

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	Paul Johannet, MD	_____	03-Apr-2026				
Clinical Operations Lead	Merck Clinical Lead	_____	03-Apr-2026				
Lead Statistician	[Name]	_____	03-Apr-2026				